Omar Hassan | 202301329

Library Management System

Question 1 — Requirements

20 Functional Requirements

ID	Requirement	Priority	Justification
F01	Add new members	Must	Members need to register to borrow books.
F02	Save member details	Must	Store names, emails, and phone numbers.
F03	Edit member details	Must	Members may change their information.
F04	Add new books	Must	The library needs to save new books.
F05	Edit book details	Must	Book information may need changes.
F06	Add book categories	Must	Categories help organize books.
F07	Add book copies	Must	One book can have many copies.
F08	Check book availability	Must	Staff need to know if a copy is available.
F09	Search for books	Must	Members can find books easily.
F10	Save librarian details	Must	The system needs to store staff information.
F11	Record book loans	Must	The system needs to know who borrowed each copy.
F12	Save loan dates	Must	Staff need to know when books are due.
F13	Record book returns	Must	Staff need to know when books are returned.
F14	Prevent double borrowing	Must	The same copy cannot be borrowed twice at once.
F15	Show borrowing history	Should	Members can see books they borrowed before.
F16	Check overdue books	Should	Staff need to know which books are late.
F17	Add fines	Should	Late returns may have fines.
F18	Record fine payments	Should	Staff need to track paid fines.
F19	Reserve books	Could	Members can reserve books they want.
F20	Check reservation status	Could	Members can see if their reservation is active.

5 Non-Functional Requirements

ID	Requirement	Priority	Justification
NF01	Keep data correct	Must	Prevent wrong or duplicate records.
NF02	Protect user information	Must	Only allowed staff can access private data.
NF03	Back up data	Must	Important data should not be lost.
NF04	Make the system fast	Should	Users should get results quickly.
NF05	Support more users and books	Should	The system should work as the library grows.